



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2023-24
MONDAY TEST

CLASS: X
SUBJECT: BIOLOGY

FULL MARKS: 40
DATE: 19.6.23

General Instructions:

- The paper consists of two printed pages.
- Answers should be to the point.
- Copy the question number carefully before answering the questions.
- Marks will be deducted for spelling errors.

1. Choose the correct answer: [5]
 - a. Karyokinesis is the division of:
 - i. Cytoplasm, ii. Nucleus, iii. Chromosome, iv. Nucleolus.
 - b. Main nitrogenous waste present in urine is:
 - i. Uric acid, ii. Creatinine, iii. Urea, iv. Amino acids.
 - c. Ultrafiltration occurs in:
 - i. Bowman's Capsule, ii. Loop of Henle, iii. DCT, iv. PCT.
 - d. The Basal Metabolic Rate in the body cells is regulated by:
 - i. Parathyroid, ii. Adrenal gland, iii. Pancreas, iv. Thyroid.
 - e. The region where crossing over takes place is known as:
 - i. Spindle fibre, ii. Centromere, iii. Chiasmata, iv. Synapsis.
2. Choose the odd one out and write the correct category to which the others belong: [5]
 - a. Mitosis, Karyokinesis, Cytokinesis, Chromosome.
 - b. Afferent arteriole, Efferent arteriole, Glomerulus, Pelvis.
 - c. Kidney, Ureter, Urethra, Uterus.
 - d. Chromatid, Chromosome, Chromatin thread, Centriole.
 - e. Goitre, Cretinism, Scurvy, Myxodema.
3. State the exact location of the following: [5]
 - a. Centromere
 - b. Pituitary gland
 - c. Thyroid gland
 - d. Kidney
 - e. Urinary bladder

4. Match the correct answers of Column I with Column II. Please rewrite the correct pairs: [5]

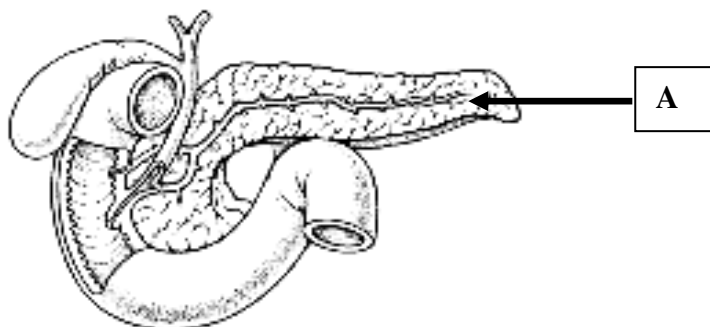
Column I	Column II
i. Bowman's Capsule	Pigment present in urine
ii. Meiosis	Blood in urine
iii. Haematuria	Cortex of kidney
iv. ADH	Chromosome number becomes half
v. Urochrome	Reabsorption of water
	Anterior lobe of pituitary gland
	Glucose in urine

5. Answer the following questions:

- Define nephron. [1]
- Differentiate between Histone and Nucleosome on the basis of their definition. [2]
- Why do we urinate more in winters than in summers? [2]
- Mention the two important features of Synthesis phase. [2]
- Draw a neat and well labelled diagram of late Anaphase. [3]

6. Answer the following questions:

- Define Tropic hormone. [1]
- Mention two important features of Metaphase. [2]
- What causes Ultrafiltration to take place in Glomerulus? [2]
- Name the gland from which we get cortisone. What do you understand by Adrenal virilism? [2]
- Observe the diagram given below and answer the following questions: [3]



- Identify the organ 'A'.
- Name the group of cells which are known as hormone secreting cells of "A".
- Mention the function of the hormone which is secreted from Beta cells of the above marked organ 'A'.